HOMIGO CARE

Because Care Begins at Home

AI Build Kit

Complete blueprint for the native Android apps, Admin panel, website and backend

Islamabad & Rawalpindi · Version 1.0 · October 2026

1. How to use this kit with an AI

HomiGo Care is a healthcare-at-home platform with a **Patient app**, a **Professional app**, an **Admin panel** and a **public website**, on one backend. This kit turns the idea into instructions an AI coding assistant can follow phase by phase.

Steps

1. Give the AI the whole ZIP folder.
2. Paste `AI_MASTER_PROMPT.md` as the first message (rules, stack, kit map).
3. Paste one file from `02_PHASE_PROMPTS/` at a time, in order. Check its Definition of Done before moving on.
4. For a 5-hour hackathon demo, use `HACKATHON_5H.md` with the mock API instead.

What is in the ZIP

Folder	Contents
`01_MASTER_GUIDE`	This PDF
`02_PHASE_PROMPTS`	12 phase prompts plus the hackathon plan
`03_SPECS`	76 screens, OpenAPI (105 operations), PostgreSQL schema and seed, roles, state machines, security, SOS, QA, demo script
`04_DESIGN_SYSTEM`	Tokens, Android Compose theme, 107 icons, official logo assets (transparent), illustrations
`05_STARTER`	Working mock API (Node 22) for developing UIs before the backend exists

Urdu summary / اردو خلاصہ

یہ کٹ ہوم گو کیئر کی مکمل بلیو پرنٹ ہے۔ اسے کسی بھی AI کوڈنگ ٹول کو دیں: پہلے **AI_MASTER_PROMPT.md** پیسٹ کریں، پھر **PHASE_PROMPTS_02** سے ایک ایک فیز ترتیب سے دیں۔ ہر فیز کے آخر میں مکمل ہونے کی فہرست دیکھیں۔ نیٹو اینڈرائیڈ ایپس (مریض اور پروفیشنل)، ایڈمن پینل، ویب سائٹ اور بیک اینڈ الگ الگ مراحل میں بنیں گے۔ آفیشل لوگو شفاف بیک گراؤنڈ کے ساتھ ہر جگہ استعمال ہوگا، کوئی ایموجی نہیں، اور تمام رنگ ہر جگہ ایک جیسے رہیں گے۔ ہیکاتھون کے لیے پانچ گھنٹے کا الگ منصوبہ شامل ہے: چار ہندسوں کا پن لاک، آل اِن ون بکنگ، اور ایڈمن ڈیش بورڈ پر سرخ SOS الرٹ۔

Demo accounts (development only)

Role	Login	Secret
Patient	+923000000002	Demo@1234, OTP 123456, Health Passport PIN 1234
Professional	+923000000003	Demo@1234, OTP 123456
Admin	admin@homigocare.test	Demo@1234, 2FA 123456

2. Product vision and roles

Mission: *Because Care Begins at Home.* Verified doctors and nurses, medicines and lab tests, delivered to the door in Islamabad and Rawalpindi.

Pillars: verified onboarding (phone, CNIC, emergency contact) · all-in-one booking (doctor appointment, nurse home visit, medicine order, lab test) · Digital Health Passport with 4-digit PIN · live GPS and SOS · admin verification and monitoring.

Hackathon MVP (5 hours): PIN lock, All-in-One Booking UI, SOS alert shown as a prominent red alert on the Admin dashboard.

29 roles and rollout

#	Role	Side	Phase	What they do
1	Individual Patient	patient	1	Books services for self
2	Family Account Manager	patient	1	Manages bookings and records for several family members
3	Home Visit Nurse (RN/BSN)	professional	1	Short visits: injections, wound care, drips, catheter care
4	Dedicated Nurse (RN/BSN)	professional	1	8/12/24 hour continuous care
5	Hospital Companion Nurse	professional	1	Stays with patient inside hospital
6	Student Nurse	professional	1	Basic tasks only under supervisor approval; labeled Student
7	Intern Nurse	professional	1	Mandatory internship; supervised; labeled Intern
8	CNA - Certified Nursing Assistant	professional	1	Basic care only: hygiene, feeding, mobility, vitals. No clinical procedures
9	CMW - Community Midwife	professional	1	Antenatal/postnatal, normal delivery support, newborn care, family planning
10	Lady Health Visitor (LHV)	professional	1	Mother and child health, vaccinations, growth monitoring
11	MLT - Medical Lab Technologist	professional	1	Home sample collection; results uploaded to Health Passport
12	Physiotherapist	professional	1	Home physiotherapy
13	Occupational Therapist	professional	1	Daily-task recovery, adaptive equipment training
14	General Physician (GP)	professional	1	Home visits for general complaints and prescriptions
15	Specialist Doctor	professional	1	Online video/audio consultations only
16	Speech Therapist	professional	2	Stroke/surgery recovery, speech and swallowing
17	Respiratory Therapist	professional	3	Oxygen, ventilator support, nebulizer
18	Psychologist / Mental Health Counselor	professional	2	Online or home therapy sessions
19	Nutritionist / Dietitian	professional	3	Diet plans for diabetes, weight, post-surgery, pregnancy
20	Palliative / Hospice Care Nurse	professional	3	Comfort care for terminally ill patients

21	Vaccination Specialist	professional	3	Home vaccination, cold-chain handling
22	Home Health Aide / Caregiver (non-medical)	professional	2	Companionship, daily activity help, mobility. No medical certificate
23	Ambulance / Emergency Driver	professional	3	Optional in-house emergency transport (alternative to Rescue 1122 / Edhi)
24	Lab Partner	partner	3	Receives orders, sends technician, uploads results
25	Pharmacy Partner	partner	3	Receives and delivers medicine orders
26	Admin	internal	1	Full access: verification, disputes, monitoring, SOS
27	Verification Officer	internal	3	Document verification only; no settings or disputes
28	Field Quality Supervisor	internal	3	Random visit checks, service quality, on-ground disputes
29	Customer Support Agent	internal	3	In-app chat/call support; no verification access

3. Decisions and ID crosswalk

#	Topic	Decision	Why
D1	Mobile tech	**Native Android (Kotlin + Compose)**. Not React Native.	Owner asked for original native apps. Better GPS, background SMS fallback, performance.
D2	Number of apps	Two apps (Patient, Professional) sharing core modules	Different audiences, store listings and permissions
D3	Backend	Node 22 + TypeScript + Express + Prisma + PostgreSQL 16	One language across API, admin, website
D4	Admin	Separate React web app, never public	Security; ops use
D5	Screen IDs	76 canonical IDs: P-01..P-33, PR-01..PR-24, A-01..A-13 (+ A-01A, A-03A, A-04A, A-05A, A-06A, A-07A)	Merges the Developer Guide, interactive master (67) and flow charts (9 extended)
D6	Phases	1 = MVP, 2 = core, 3 = full. `showcase` flag marks the hackathon set	Ships value early
D7	Roles	29 roles; Phase 1 = roles 1-15 + Admin; Phase 2 = Speech Therapist, Psychologist, Home Health Aide; Phase 3 = rest	Matches Roles Update doc
D8	Payments	Provider interface; `SANDBOX` first; JazzCash/Easypaisa later. 15% professional commission	No live merchant credentials yet
D9	OTP	Provider interface; dev code 123456 when `OTP_DEV_BYPASS=true`	Testing without SMS cost
D10	SOS	Honest SOS: real alert + real device location to Admin. No fake dispatch. Offline = SMS fallback where the device allows	Safety and trust
D11	Hackathon	A "showcase" track inside Phase 1 delivers PIN lock, All-in-One Booking UI, SOS alert in 5 hours using the mock API	Matches hackathon rules
D12	Logo	Official raster art supplied by owner; all launcher, store, web and OG assets derived from it	Brand consistency
D13	Logo vs UI color	Logo turquoise differs from UI teal `#1B6B63`. UI uses brand tokens; logo accents (`#28B8C0` `#0C6A80` `#F87848` `#F5B82E`) only in logo/illustration	Contrast + consistency
D14	Icons	Original 107-icon set, 24 grid, 1.75 stroke, round caps. No emoji	Professional look
D15	Languages	EN + UR, RTL aware	Audience
D16	Admin sees medical records?	No. Admin sees booking metadata and verification documents only	Privacy
D17	Assessment	3 phases (70/75/80% pass; 30/20/20 questions), 3 attempts, 7-day cooldown. Questions are SAMPLES	Per Developer Guide
D18	Health Passport	Patient grants access per booking; PIN 4 digits, lockout after 5 wrong tries for 15 min; every read logged immutably	Privacy

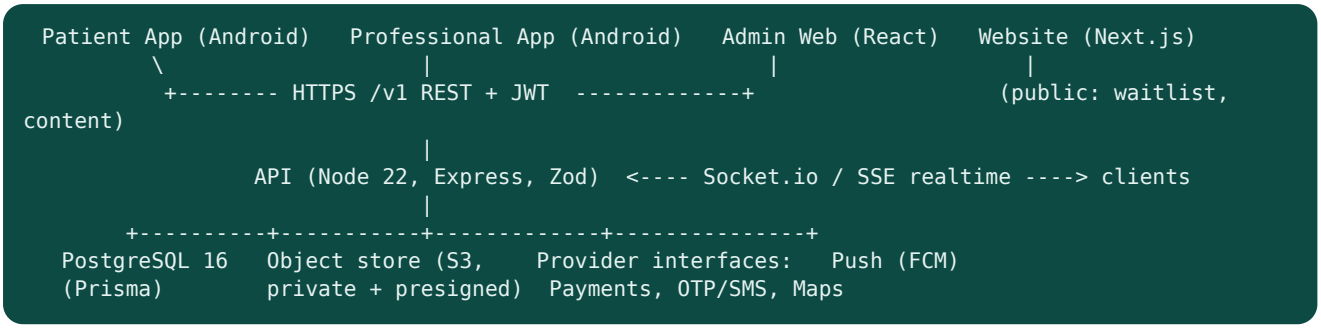
ID crosswalk

- Developer Guide v1.0/1.1 numbering `P-01..` and `PR-..`, `A-..` is preserved.
- Interactive Spider-Web master (67 screens) maps 1:1 to the same IDs.
- Screens flagged `status: EXTENDED` in `screens.json` come from the flow-chart images and were not in the original guide; they are small additions (for example sub-screens A-03A, A-04A).
- If two sources disagree on a label, `screens.json` wins.

Hackathon vs phase flags

Three showcase screens carry later phase numbers in the source guide (P-16 and P-24 Health Passport = phase 2, P-25 SOS = phase 3). For the 5-hour hackathon they are pulled forward (`showcase: true`). In the full build, implement them in Phase 07 (`07_HEALTH_PASSPORT_SOS`) regardless of the number. Key IDs: dashboard A-02, SOS monitor A-06, alert detail A-06A, verification A-03/A-03A.

4. Architecture and stack



Repo layout (monorepo, pnpm workspaces)

```
apps/android/            Gradle multi-module
  app-patient/  app-pro/  core-designsystem/  core-network/  core-data/  core-ui/  feature-*/
apps/admin-web/          React + Vite
apps/website/            Next.js 15
services/api/            Express + Prisma
packages/design-tokens/  tokens.json, tokens.css, tailwind preset (copied from 04_DESIGN_SYSTEM)
packages/api-types/      generated from openapi.yaml
```

Layers (backend)

routes → controller (Zod validation) → service (business rules, state machines) → repository (Prisma). Providers (payments, OTP, push, storage) are interfaces with a sandbox implementation first.

Layers (Android)

UI (Compose screens, stateless) → ViewModel (StateFlow, UiState sealed class: Loading/Content/Empty/Error) → Repository → (Retrofit API | Room cache). Hilt for DI. Every screen implements all four states.

Realtime

Events (see `x-realtime` in openapi.yaml): `booking.created`, `booking.updated`, `sos.created`, `sos.updated`, `professional.verification`, `notification.created`. Socket.io in production; the mock uses SSE at `/v1/rt/stream?token=`. Clients must reconnect with backoff and refetch on reconnect.

Environments

Env	Purpose	Notes
local	dev	docker compose: postgres + api; mock-api optional
staging	QA, demo	OTP_DEV_BYPASS may stay on; sandbox payments
production	live	bypass OFF, Admin behind VPN/IP allowlist + 2FA, backups, TLS

Environment variables (services/api/.env.example)

`DATABASE_URL, JWT_ACCESS_SECRET, JWT_REFRESH_SECRET, FIELD_ENC_KEY (32 bytes base64), S3_BUCKET, S3_REGION, S3_ACCESS_KEY, S3_SECRET, FCM_SERVICE_ACCOUNT_JSON, OTP_PROVIDER, OTP_DEV_BYPASS, PAYMENT_PROVIDER, CORS_ORIGINS, ADMIN_ALLOWED_IPS`

Performance targets

API p95 < 400 ms for list endpoints; app cold start < 2.5 s on a mid-range device; SOS create round trip < 2 s on 4G.

Stack summary

Part	Technology
Android (Patient, Professional)	Kotlin, Jetpack Compose, Material3 themed, Hilt, Retrofit/OkHttp, Room, Navigation Compose, FCM, Maps Compose, minSdk 26
Admin web	React, Vite, TypeScript, Tailwind, TanStack Query
Website	Next.js 15, TypeScript, Tailwind, EN/UR with RTL
Backend	Node 22, TypeScript, Express, Zod, Prisma, PostgreSQL 16, Socket.io, JWT
Security	bcrypt 12+, AES-256-GCM fields, S3 presigned URLs, idempotency keys, append-only audit

5. Design system

Official logo (transparent)



Mark



Horizontal



Stacked



Horizontal (white)



Stacked (white)

Launcher icon and store graphics



Launcher



Round



Play feature graphic



OG image

Folder `04_DESIGN_SYSTEM/logo_assets/` (transparent PNGs):

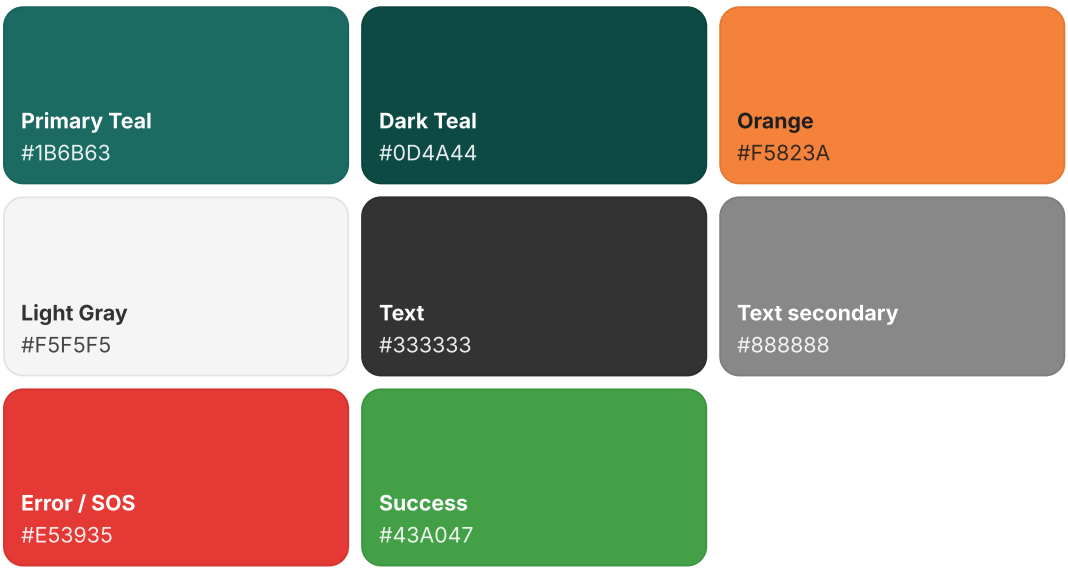
File	Use
`homigo_mark.png`	In-app header marks, empty states, loading
`homigo_lockup_horizontal(.png/_white.png)`	App bars on wide screens, website header, admin sidebar. `_white` on teal/dark backgrounds
`homigo_lockup_stacked(.png/_white.png)`	Splash, login, onboarding, PDF cover
`homigo_wordmark*.png` , `homigo_tagline*.png`	Only when composing custom lockups
`homigo_mark_on_white_tile.png`	Places that need an opaque square
`android/mipmap-*/ic_launcher*.png` , `mipmap-anydpi-v26/*.xml`	Copy into `app-patient/src/main/res` and `app-pro/src/main/res`

File	Use
`play_store_icon_512.png`, `play_feature_graphic_1024x500.png`	Play Console
`web/favicon-*.png`, `favicon.ico`, `og_image_1200x630.png`	Website and admin `<head>`

Rules: never recolor, redraw, stretch, rotate, add shadows or outlines. Keep clear space equal to 25% of the mark height. Minimum mark size 32 dp; below that use the favicon file. The mark must not sit on busy imagery; use the white tile or a plain surface. Splash: stacked lockup on white (or `_white` on teal gradient).

Why raster: the owner supplied raster 3D art. Use density-specific files; do not upscale. A vector master is recommended for the future.

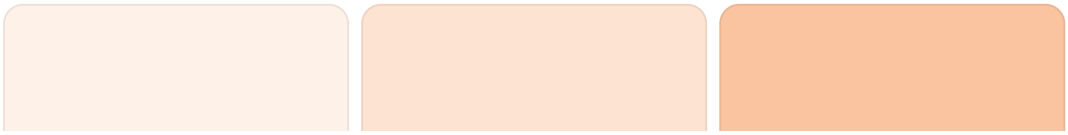
Brand colors



Teal scale



Orange scale



50
#FEF1E8

100
#FDE3D1

200
#FAC4A0

500
#F5823A

700
#A8501A

Logo accents (logo and illustration only)

logoCyan
#28B8C0

logoTealDeep
#0C6A80

logoOrange
#F87848

logoYellow
#F5B82E

Contrast checks

Foreground	Background	Use	Ratio	AA text
#FFFFFF	#1B6B63	Button text on primary	6.31	Pass
#FFFFFF	#0D4A44	Text on dark teal	10.1	Pass
#1B6B63	#FFFFFF	Teal text on white	6.31	Pass
#333333	#FFFFFF	Body text	12.63	Pass
#888888	#FFFFFF	Secondary text (#888)	3.54	Large only
#666666	#FFFFFF	Secondary text AA (#666)	5.74	Pass
#FFFFFF	#F5823A	White on orange	2.58	Fail
#0D4A44	#F5823A	Dark teal on orange	3.91	Large only
#FFFFFF	#E53935	White on SOS red	4.23	Large only
#FFFFFF	#43A047	White on success green	3.3	Large only
#F5823A	#FFFFFF	Orange text on white	2.58	Fail
#A8501A	#FFFFFF	Orange-700 text on white (use for orange text)	5.49	Pass
#2D7A31	#FFFFFF	Green-700 text on white	5.33	Pass
#A52824	#FFFFFF	Red-700 text on white	7.17	Pass

Shape and spacing

Buttons and inputs radius 25, cards 15, chips 20, sheets 24 (top). 4-pt grid: 4, 8, 12, 16, 24, 32. Screen gutter 16. Minimum touch target 48 dp.

Type

Brand 28 bold, H1 22 bold, H2 18 semibold, body 14, small 12, button 16 bold. Latin: Inter (bundle files; fall back to system). Urdu: Noto Nastaliq Urdu (bundle files; the kit could not include font binaries, download from

Google Fonts).

Status badges

Pending · Under review · Approved · Rejected · Action required · Accepted · On the way · Arrived · In progress · Completed · Cancelled · Declined · Active SOS · Acknowledged · Resolved · Online · Offline · Suspended

Icon set (107 original icons, 24 grid, 1.75 stroke)



wifi-off

x

x-circle

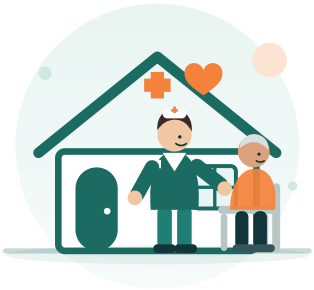
Illustrations



certification



empty_state



onboarding_1_home_care



onboarding_2_verified



onboarding_3_safety



under_review

6. Screen catalog (76 screens)

Phase 1 = MVP, 2 = core, 3 = full. **Showcase** marks the 5-hour hackathon set. Full details (API calls, components, states, behavior) are in `03_SPECS/screens/`.

Patient app (33)

ID	Screen	Purpose	Ph	Next	API
P-01	Splash *	Launch and brand entry	1	P-02	GET /me (if token)
P-02	Onboarding 1 *	Introductory onboarding	1	P-03	
P-03	Onboarding 2 *	Explain core healthcare value	1	P-04	
P-04	Onboarding 3 *	Complete onboarding and enter authentication	1	P-05	
P-05	Login *	Patient authentication	1	P-11, P-06	POST /auth/login; POST /auth/forgot-password
P-06	Sign Up *	Create patient account	1	P-07	POST /auth/signup
P-07	OTP *	Verify phone number	1	P-08	POST /auth/verify-otp; POST /auth/resend-otp
P-08	Profile Setup 1	Basic profile information	1	P-09	PUT /patient/profile; POST /patient/documents/upload-url
P-09	Profile Setup 2 — Address	Patient address/location	1	P-10	POST /patient/addresses
P-10	Profile Setup 3 — Emergency Contact	Emergency contact setup	1	P-11	PUT /patient/emergency-contact
P-11	Home Dashboard *	Main patient control center	1	P-12, P-13, P-16, P-25, P-26, P-27, P-28, P-30, P-31, P-32	GET /catalog/services; GET /professionals?sort=rating; GET /bookings?tab=completed; GET /n
P-12	Service Category	Select healthcare service category	1	P-13, P-26, P-27	GET /catalog/services?category=NURSE
P-13	Browse Professionals	Browse doctors/nurses	1	P-14	GET /professionals; PUT /patient/favourites/{id}
P-14	Professional Profile	Review professional information	1	P-15	GET /professionals/{id}; GET /professionals/{id}/reviews; GET /professionals/{id}/availabi
P-15	Booking Details *	Enter service/visit details	1	P-16, P-17, P-18	POST /bookings/quote; GET /patient/addresses
P-16	Health Passport PIN / Access *	Authorize protected Health Passport access with 4-digit PIN	2	P-18	POST /health-passport/verify-pin; POST /health-passport/grants

P-17	Supply Pre-Request	Collect additional booking/request information when required	2	P-18	PUT /bookings/{id}/supplies/{sid}/decision
P-18	Booking Summary & Payment	Review booking and payment	1	P-19	POST /bookings; POST /payments/confirm
P-19	Booking Confirmed	Confirmation and booking reference	1	P-20, P-22	GET /bookings/{id}
P-20	Live Tracking	Track applicable home visit	2	P-21	WS booking:{id} tracking.location
P-21	Visit In Progress	Active visit state	2	P-22, P-25	WS booking:{id}
P-22	Visit Complete	Completed service	2	P-23, P-24	GET /bookings/{id}
P-23	Rating & Review	Rate professional and leave feedback	2	P-32	POST /bookings/{id}/review
P-24	Health Passport *	View digital medical record after authorization	2	P-11, P-16	GET /health-passport; GET /health-passport/records; GET /health-passport/access-log
P-25	Emergency Screen / SOS *	Trigger emergency alert	3	A-06	POST /sos; POST /sos/{id}/location; POST /sos/{id}/cancel
P-26	Lab Test Booking	Select and schedule lab test	3	P-18	GET /catalog/services?category=LAB_TEST
P-27	Medicine Order	Search/upload prescription and place medicine order	3	P-18	POST medicine order (phase 3)
P-28	Wallet	View/manage wallet information	3	P-11	GET /wallet
P-29	Referral	Referral feature	3	P-11	GET /referrals
P-30	Notifications	Patient notifications	2	P-11, P-19, P-25	GET /notifications; POST /notifications/read-all
P-31	Profile & Settings	Manage patient account/settings	3	P-11	GET /me; PUT /me; DELETE /me
P-32	Booking History	View previous bookings/orders	2	P-11, P-14	GET /bookings?tab=
P-33	Help & Support	FAQs, contact support, raise a ticket	3	P-31	POST /public/contact

Professional app (24)

ID	Screen	Purpose	Ph	Next	API
PR-01	Professional Splash	Professional entry	1	PR-02	
PR-02	Professional Login / Role Entry	Authenticate and choose professional path	1	PR-08, PR-03	POST /auth/login

PR-03	Role & Registration	Doctor/nurse registration	1	PR-04	POST /auth/signup; POST /auth/verify-otp
PR-04	Personal & Professional Details	Professional profile details	1	PR-05	POST /professional/register
PR-05	Credential & Document Upload	Upload degree/certification/identity documents	1	PR-06	POST /professional/documents/upload-url; POST /professional/documents
PR-06	Verification Review / Submission	Review and submit credentials	1	PR-07	POST /professional/submit-verification
PR-07	Verification Pending / Status	Show verification state	1	PR-09, PR-08	GET /professional/verification-status; WS verification.updated
PR-08	Professional Dashboard *	Main doctor/nurse control center	1	PR-11, PR-17, PR-18, PR-19, PR-10, PR-20	GET /professional/dashboard; PUT /professional/online; WS booking.offer
PR-09	Certification / Assessment	Professional assessment when required	1	PR-08	GET /professional/assessment; POST .../start; PUT .../answer; POST .../submit; POST /profe
PR-10	Professional Profile Management	Manage public professional profile	2	PR-08	GET/PUT /professional/profile
PR-11	Incoming Booking Requests	View incoming patient requests	1	PR-12	GET /bookings (offers); WS booking.offer
PR-12	Booking Request Details *	Review booking details	1	PR-13, PR-11	GET /bookings/{id}; POST /bookings/{id}/accept; POST /bookings/{id}/decline
PR-13	Booking Accepted / Upcoming *	Upcoming accepted service	1	PR-14	GET /bookings/{id}
PR-14	Active Visit / Consultation	Active consultation/home visit	2	PR-15, PR-16	PUT /bookings/{id}/status; POST /bookings/{id}/location; POST /bookings/{id}/supplies
PR-15	Authorized Patient Health Passport	Access patient passport only after authorization	2	PR-16	GET /professional/patients/{id}/health-passport
PR-16	Complete Service / Visit	Complete service and record completion	2	PR-17	POST /professional/bookings/{id}/visit-notes; PUT /bookings/{id}/status
PR-17	Professional Booking History	Previous services	2	PR-08, PR-11	GET /bookings
PR-18	Professional Notifications	Professional alerts	2	PR-08, PR-12	GET /notifications

PR-19	Availability & Service Settings	Set availability and service preferences	2	PR-08	PUT /professional/availability
PR-20	Professional Settings & Account	Manage account/security settings	3	PR-08, PR-01	PUT /me
PR-21	Earnings & Wallet	Earnings history, balance, withdraw to JazzCash/Easypaisa/bank (min PKR 500, 24 h)	2	PR-08	GET /professional/earnings; POST /wallet/withdraw
PR-22	Ratings & Reviews	Patient ratings, feedback, reputation	2	PR-08	GET /professionals/{me}/reviews
PR-23	Messages	Patient chat, admin messages, support	3	PR-08	
PR-24	Help & Support	FAQs, contact admin, raise issue	3	PR-20	POST /public/contact

Admin panel (19)

ID	Screen	Purpose	Ph	Next	API
A-01	Admin Login *	Secure administrator authentication	1	A-02	POST /auth/login
A-01A	Admin 2FA / OTP Verification *	Second factor after password — mandatory for every admin login	1	A-02	POST /auth/admin/2fa
A-02	Admin Dashboard *	Administrative command center	1	A-03, A-04, A-05, A-06, A-07, A-08, A-09, A-10	GET /admin/stats; GET /admin/professionals?status=pending; GET /admin/sos?status=open; WS
A-03	Professional Verification *	Review professional verification queue	1	A-03A	GET /admin/professionals
A-03A	Professional Verification Detail *	Review credentials and decide verification	1	A-03, A-09	GET /admin/professionals/{id}; PUT /admin/professionals/{id}/decision
A-04	User Management	Manage patient accounts	1	A-04A	GET /admin/users
A-04A	User Detail	Review account-level details	2	A-04, A-05A, A-07A	GET /admin/users/{id}; PUT /admin/users/{id}/status
A-05	Booking Management *	Monitor bookings	1	A-05A	GET /admin/bookings
A-05A	Booking Detail	Review booking operational details	2	A-05, A-07A	GET /admin/bookings/{id}
A-06	Emergency / SOS Monitoring *	Monitor emergency queue	1	A-06A	GET /admin/sos; WS sos.created / sos.updated

A-06A	Emergency Alert Detail *	Acknowledge and manage SOS response	1	A-06	GET /admin/sos/{id}; POST .../acknowledge; POST .../events; POST .../resolve
A-07	Dispute & Issue Management	Manage issues/disputes	2	A-07A	GET /admin/disputes
A-07A	Issue / Dispute Detail	Resolve or escalate issue	2	A-07	GET/PUT /admin/disputes/{id}; POST .../notes
A-08	Platform Monitoring & Statistics	Platform metrics and statistics	2	A-02	GET /admin/stats
A-09	Admin Notifications	Operational alerts	1	A-02, A-03A, A-06A	GET /admin/notifications
A-10	Admin Profile & Settings	Admin account/settings	2	A-02, A-01	GET /me; PUT /me
A-11	Audit Log & Activity	Immutable list of every admin action (flow chart rule: all actions are logged)	2	A-02	GET /admin/audit-logs
A-12	Help & Support / Knowledge Base	FAQs, contact support, raise a ticket (flow chart A-15)	3	A-02	POST /public/contact
A-13	Broadcast & Message Templates	Send broadcast notifications and manage SMS/email/push templates (flow chart A-11/A-13)	3	A-09	GET/PUT /admin/settings

Cross-panel flows

From	To	Event
P-25	A-06	PATIENT_SOS_TO_ADMIN
P-19	PR-11	PATIENT_BOOKING_TO_PROFESSIONAL
PR-13	P-19	PROFESSIONAL_ACCEPTS_PATIENT_BOOKING
PR-14	P-21	PROFESSIONAL_ACTIVE_VISIT_TO_PATIENT
P-16	PR-15	PATIENT_AUTHORIZES_HEALTH_PASSPORT
P-25	A-06A	SOS_ALERT_DETAIL
PR-05	A-03	PROFESSIONAL_CREDENTIALS_TO_ADMIN_VERIFICATION
A-03A	PR-07	ADMIN_VERIFICATION_RESULT_TO_PROFESSIONAL_STATUS

* marks showcase screens.

7. State machines

Booking

```
PENDING -> ACCEPTED -> ON_THE_WAY -> ARRIVED -> IN_PROGRESS -> COMPLETED
PENDING -> DECLINED | EXPIRED | CANCELLED
ACCEPTED/ON_THE_WAY -> CANCELLED (fee rules apply)
```

- Only the assigned professional may move ACCEPTED→…→COMPLETED. Only the patient or admin may CANCEL.
- Accept is atomic: first professional wins; second gets 409 `ALREADY_TAKEN`.
- INSTANT bookings expire after 5 minutes without acceptance → EXPIRED and patient is notified.
- Create and payment endpoints require an `Idempotency-Key` header.
- Completion triggers: payment capture (SANDBOX), commission row (15%), rating prompt, receipt.

Professional verification

```
DRAFT -> SUBMITTED -> UNDER_REVIEW -> APPROVED
                                   -> REJECTED (reason required)
                                   -> ACTION_REQUIRED (list of docs to fix) -> SUBMITTED
APPROVED -> SUSPENDED (admin, reason required) -> UNDER_REVIEW
```

Only APPROVED professionals appear in search and can go online. Every decision writes `audit_logs`.

Assessment

3 phases in order. Phase 1: 30 questions, pass 70%. Phase 2: 20 questions, pass 75%. Phase 3: 20 questions, pass 80%. Max 3 attempts per phase; after the 3rd failure, 7-day cooldown. Passing all three is required before submission can move to UNDER_REVIEW.

SOS

```
ACTIVE -> ACKNOWLEDGED -> RESPONDING -> RESOLVED
ACTIVE/ACKNOWLEDGED/RESPONDING -> ESCALATED -> RESOLVED
ACTIVE/ACKNOWLEDGED -> FALSE_ALARM
```

Each transition appends to the SOS timeline with actor and time.

Health Passport access

`NONE -> GRANTED(booking, expires at booking end + 2h) -> EXPIRED | REVOKED`. A professional may read only while GRANTED. Each read appends to `access_logs` (append-only; DB trigger blocks update/delete).

Payment

`INITIATED -> SUCCESS | FAILED`, `SUCCESS -> REFUNDED`. Never mark SUCCESS from the client; only from the provider callback or the sandbox confirm endpoint.

8. Data model

PostgreSQL 16, 43 tables, enums for every status, append-only triggers on `access_logs` and `audit_logs`, view `v_admin_kpis`. Files: `03_SPECS/database/schema.sql`, `seed.sql`.

`users` `otp_codes` `refresh_tokens` `devices` `admin_totp` `patients` `emergency_contacts` `addresses` `documents` `professionals` `verification_events` `availability_rules` `assessment_questions` `assessment_attempts` `services` `professional_services` `bookings` `booking_offers` `booking_events` `booking_checklist_items` `supply_requests` `payments` `wallets` `wallet_transactions` `reviews` `health_passports` `health_records` `vitals` `access_grants` `access_logs` `sos_alerts` `sos_events` `disputes` `dispute_notes` `notifications` `audit_logs` `referrals` `partners` `lab_tests` `lab_orders` `medicine_orders` `feature_flags` `waitlist_signups`

9. API summary

Base `/v1`. Contract: `03_SPECS/api/openapi.yaml` (OpenAPI 3.0.3). Error shape `{error: {code,message,fields}}`. Create and payment calls need `Idempotency-Key`.

Auth (12)

Method	Path	Summary
POST	`/auth/signup`	Create account (patient or professional) and send OTP
POST	`/auth/verify-otp`	Verify 6-digit OTP (expires 10 min, 3 attempts)
POST	`/auth/resend-otp`	Resend OTP (after 60 s)
POST	`/auth/login`	Phone/email + password. Admin login returns requires2fa=true
POST	`/auth/admin/2fa`	Complete admin 2FA
POST	`/auth/refresh`	Rotate refresh token
POST	`/auth/forgot-password`	Start OTP password reset
POST	`/auth/reset-password`	Set new password with verified OTP
POST	`/auth/logout`	Revoke refresh token
POST	`/devices`	Register FCM token
GET	`/me`	Current user
DELETE	`/me`	Request account deletion (OTP confirmed

Patient (12)

Method	Path	Summary
GET	`/patient/profile`	Get patient profile (P-08..P-10
PUT	`/patient/profile`	Update personal info (CNIC
POST	`/patient/documents/upload-url`	Get short-lived presigned PUT URL for CNIC photos / prescriptions
GET	`/patient/addresses`	List addresses
POST	`/patient/addresses`	Add address
PUT	`/patient/addresses/{id}`	Update address
DELETE	`/patient/addresses/{id}`	Delete address
PUT	`/patient/emergency-contact`	Set emergency contact (P-10)
GET	`/patient/family`	List family profiles (Phase 3)
POST	`/patient/family`	Add family member profile (Phase 3)
PUT	`/patient/favourites/{professionalId}`	Save favourite
DELETE	`/patient/favourites/{professionalId}`	Remove favourite

Catalog (5)

Method	Path	Summary
GET	`/catalog/services`	Service categories and types (P-11

GET	`/professionals`	Browse verified, live professionals (P-13)
GET	`/professionals/{id}`	Public professional profile (P-14)
GET	`/professionals/{id}/reviews`	Reviews
GET	`/professionals/{id}/availability`	Bookable slots for a date range

Professional (17)

Method	Path	Summary
POST	`/professional/register`	Set role/type + personal details (PR-03
GET	`/professional/documents`	My documents + review status
POST	`/professional/documents`	Register an uploaded document (after presigned PUT)
POST	`/professional/documents/upload-url`	Presigned PUT URL
POST	`/professional/submit-verification`	Submit for admin review (PR-06) — DRAFT/ACTION_REQUIRED → SUBMITTED
GET	`/professional/verification-status`	Status timeline (PR-07)
GET	`/professional/assessment`	Phase cards + lock state (PR-09)
POST	`/professional/assessment/{phase}/start`	Start attempt; server picks questions and enforces timer
PUT	`/professional/assessment/attempts/{id}/answer`	Auto-save one answer
POST	`/professional/assessment/attempts/{id}/submit`	Submit (also called by server on deadline); returns score and next unlock
GET	`/professional/profile`	Own profile (PR-10)
PUT	`/professional/profile`	Update bio
POST	`/professional/go-live`	Activate profile (PR-09 end) — only if APPROVED and assessment complete
PUT	`/professional/availability`	Weekly availability rules (PR-19)
PUT	`/professional/online`	Toggle online/offline
GET	`/professional/dashboard`	Today's earnings
GET	`/professional/earnings`	Earnings summary and history (Phase 2)

Bookings (13)

Method	Path	Summary
POST	`/bookings`	Create booking (P-15 → P-18). Requires Idempotency-Key header.
GET	`/bookings`	My bookings (patient) or assigned (professional)
POST	`/bookings/quote`	Price breakdown preview (visit fee
GET	`/bookings/{id}`	Booking detail
POST	`/bookings/{id}/accept`	Professional accepts offer (PR-12 → PR-13). First valid accept wins.
POST	`/bookings/{id}/decline`	Professional declines; booking re-offered to next professional

PUT	`/bookings/{id}/status`	Professional advances: ON_THE_WAY → ARRIVED → IN_PROGRESS → COMPLETED (Phase 2)
POST	`/bookings/{id}/cancel`	Cancel (patient or admin)
POST	`/bookings/{id}/location`	Professional posts GPS ping (every 3 s via socket preferred)
POST	`/bookings/{id}/supplies`	Professional requests supplies (P-17)
PUT	`/bookings/{id}/supplies/{supplyId}/decision`	Patient approves / requests changes / declines
POST	`/bookings/{id}/review`	Rate & review (P-23); credits PKR 50 cashback
POST	`/bookings/{id}/dispute`	Report an issue

Payments (5)

Method	Path	Summary
POST	`/payments/confirm`	Confirm payment for a booking (P-18 → P-19). Booking leaves PENDING-offer state only after SUCCESS.
GET	`/wallet`	Balance + last transactions (P-28
POST	`/wallet/topup`	Add money (Phase 3)
POST	`/wallet/withdraw`	Withdraw (professional; min PKR 500)
GET	`/referrals`	Code

HealthPassport (10)

Method	Path	Summary
GET	`/health-passport`	Patient's own passport summary (P-24). Requires recent biometric/PIN re-auth header X-Passport-Token.
PUT	`/health-passport/pin`	Create or change 4-digit PIN (bcrypt + pepper server-side)
POST	`/health-passport/verify-pin`	Patient proves PIN; returns short-lived X-Passport-Token (5 min). 5 failures ⇒ lockout with backoff.
POST	`/health-passport/grants`	Patient authorises a professional for a booking (P-16); scope + expiry
DELETE	`/health-passport/grants/{id}`	Revoke
GET	`/health-passport/access-log`	Immutable access log (P-24 tab)
GET	`/health-passport/records`	History
POST	`/health-passport/records`	Add own record/document
GET	`/professional/patients/{patientId}/health-passport`	Professional reads authorised scope ONLY (PR-15). 403 + DENIED log without a valid grant.
POST	`/professional/bookings/{id}/visit-notes`	Professional writes visit note / prescription into the patient's record (PR-16)

SOS (4)

Method	Path	Summary
POST	`/sos`	Trigger SOS (P-25). Patient or professional. Admin sees it instantly via socket.

POST	`/sos/{id}/location`	Append location update
POST	`/sos/{id}/cancel`	Triggerer marks false alarm
POST	`/sos/sms-relay`	Webhook for the offline-SMS relay number; creates SMS_RELAY alert (Phase 3)

Notifications (3)

Method	Path	Summary
GET	`/notifications`	List (30 days)
POST	`/notifications/read-all`	Mark all read
POST	`/notifications/{id}/read`	Mark read

Admin (22)

Method	Path	Summary
GET	`/admin/stats`	KPI cards + series (A-02
GET	`/admin/professionals`	Verification queue (A-03)
GET	`/admin/professionals/{id}`	Detail + documents (signed URLs
PUT	`/admin/professionals/{id}/decision`	Approve / Reject / Request changes — reason mandatory for reject/changes; writes verification_events + audit_logs; notifies professional
GET	`/admin/users`	Patients/accounts (A-04)
GET	`/admin/users/{id}`	Account detail — NO medical records (A-04A)
PUT	`/admin/users/{id}/status`	Suspend / reactivate (audited
GET	`/admin/bookings`	All bookings (A-05)
GET	`/admin/bookings/{id}`	Booking detail + timeline (A-05A)
GET	`/admin/sos`	Active first
GET	`/admin/sos/{id}`	Alert detail + timeline (A-06A)
POST	`/admin/sos/{id}/acknowledge`	Acknowledge
POST	`/admin/sos/{id}/events`	Add note / contact attempt / escalate / update response status
POST	`/admin/sos/{id}/resolve`	Resolve or mark false alarm (note required)
GET	`/admin/disputes`	Issue queue (A-07)
GET	`/admin/disputes/{id}`	Detail (A-07A)
PUT	`/admin/disputes/{id}`	Assign / change status / resolve / escalate
POST	`/admin/disputes/{id}/notes`	Add internal note
GET	`/admin/notifications`	Alert stream grouped by emergency/verification/booking/issue/system (A-09)
GET	`/admin/audit-logs`	Immutable audit trail (A-11)
GET	`/admin/settings`	Feature flags
PUT	`/admin/settings`	Update (super admin only)

Public (2)

Method	Path	Summary

POST	`/public/waitlist`	Website waitlist / professional interest form (rate limited
POST	`/public/contact`	Contact form

10. Security and privacy

Auth

- Phone + OTP verification at signup. Password hashed with bcrypt cost >= 12.
- JWT access token 15 min; refresh token rotating, stored hashed, revoked on reuse (reuse = revoke whole family).
- Admin: email + password + 2FA (TOTP in production). Admin app only reachable through VPN / IP allowlist.
- Rate limits: login 5/min/IP+account, OTP send 3/10 min/phone, PIN verify 5 tries then 15 min lock.

Data

- TLS everywhere. HSTS on web.
- Field-level AES-256-GCM encryption for CNIC and medical fields (allergies, conditions, medications, notes). Keys from env/KMS, never in code.
- Documents in a private bucket; access only through short-lived presigned URLs (5 min). Never public URLs.
- Android: tokens in EncryptedSharedPreferences/Keystore. `FLAG_SECURE` on Health Passport and document screens. Disable backup of sensitive data. Certificate pinning optional in production.

Access control

- Role checks on every route (patient/professional/admin sub-roles). Object-level checks: a patient reads only their own bookings; a professional reads only assigned bookings and only granted passports.
- Admin cannot read medical records. Admin may read verification documents. All admin actions are written to `audit_logs` (append-only).

Logging

Never log: PIN, OTP, password, tokens, CNIC, medical fields, document URLs. Use request ids. Redact by default.

Input

Zod validation on every body/query. Upload limits: 10 MB, types jpeg/png/pdf/m4a/opus, content sniffing, virus scan hook.

Compliance notes (owner must confirm with counsel)

Pakistan Personal Data Protection Bill direction, PMDC / PNC verification, DRAP for pharmacies. Provide data export and account deletion (DELETION_PENDING with 30 days grace).

OWASP mobile/web checks to run in Phase 11

Broken access control, IDOR on bookings/documents/passport, auth bypass, injection, insecure storage, exported components, WebView misuse, dependency vulnerabilities.

11. SOS emergency specification

Goal: a patient can raise an emergency alert in two taps; the Admin dashboard shows a prominent red alert instantly with the patient's name, phone, emergency contact and last known location.

Patient flow (P-25 Emergency Screen / SOS; the active state is shown on the same screen)

1. Large red SOS button (always reachable from Home). Press and hold 1.5 s OR tap then confirm (avoid accidents). Haptic feedback.
2. App captures: current location if permission granted (lat, lng, accuracy, timestamp), battery %, network type. If no GPS fix within 5 s, send with last known location and mark it "last known".
3. `POST /v1/sos` with `Idempotency-Key`. Server stores, notifies Admin (realtime + push), and notifies the patient's emergency contact by SMS (provider interface).
4. The active state of P-25 shows: "Alert sent", elapsed time, what is known ("Help team notified at 14:32"), a call button for 1122 (dialer intent), cancel as "false alarm" with confirmation. Location keeps updating through `POST /v1/sos/{id}/location` every 15 s.

Honesty rules

- Never claim an ambulance is on the way unless an admin set status RESPONDING.
- Never show fake ETAs, fake responder names or fake maps pins.
- If the alert could not be sent, say so and show the fallback (call 1122, SMS).

Offline fallback (best effort)

If there is no data connection, the Patient app prepares an SMS with a short text and a maps link to the emergency contact and to the SOS relay number (`SOS_RELAY_NUMBER`). The user confirms sending (Android does not allow silent SMS without permission; request `SEND_SMS` only if the product owner accepts the Play Store policy review, otherwise open the SMS composer). The relay posts a `SMS_RELAY` SOS to the API. This depends on device and carrier; document it as best effort.

Admin (A-06 SOS Monitoring, A-06A alert detail, dashboard A-02)

- Prominent red banner on the dashboard with sound (user-gesture permitted) and a pulsing dot. Respect `prefers-reduced-motion` (no pulse, keep color and text).
- List sorted newest first, with status chips; detail shows map, timeline, patient contact, emergency contact and actions: Acknowledge, Responding, Escalate, Resolve, Mark false alarm. Each action writes the timeline and audit log.
- Realtime events: `sos.created`, `sos.updated`.

Acceptance tests

1. Create SOS from the mock API as the patient; the admin SSE stream receives `sos.created` within 1 s.
2. Retry with the same Idempotency-Key creates one alert only.
3. Airplane mode shows the offline fallback screen with call and SMS actions.
4. Admin sees no medical records in the SOS detail.

12. Phase plan

Phase	Prompt file	Result
00	`00_FOUNDATION.md`	Create the monorepo and tooling so every later phase has a place to live.
01	`01_BACKEND_CORE.md`	Implement the API foundation from openapi.yaml and schema.sql.
02	`02_BOOKINGS_PAYMENTS_REALTIME.md`	Implement the booking lifecycle end to end.
03	`03_ANDROID_CORE.md`	Create the Gradle multi-module Android project and the design system, with the official logo.
04	`04_PATIENT_APP.md`	Build the Patient app from screens_patient.md, in the order of phase flags (phase 1 first).
05	`05_PROFESSIONAL_APP.md`	Build the Professional app.
06	`06_ADMIN_WEB.md`	Build the admin panel in React + Vite with the shared tokens and official logo.
07	`07_HEALTH_PASSPORT_SOS.md`	Finish the two safety-critical features end to end across backend, apps and admin.
08	`08_PHASE2_FEATURES.md`	Add all screens and features flagged phase 2 in screens.json and roles 16, 18, 22.
09	`09_PHASE3_FEATURES.md`	Add everything flagged phase 3 and the remaining roles.
10	`10_WEBSITE.md`	Build the marketing site in Next.js 15 with EN/UR and RTL.
11	`11_QA_SECURITY_RELEASE.md`	Prove quality and ship.

Hackathon 5-hour MVP plan

Goal: a convincing demo of three things, using the mock API so nothing blocks on the backend. 1. Digital Health Passport with a 4-digit PIN lock. 2. All-in-One Booking UI (doctor appointment, nurse home visit, medicine order, lab test). 3. SOS alert that appears as a prominent red alert on the Admin dashboard.

Use only screens with `showcase: true` in `03_SPECS/screens/screens.json`.

Timeline

Hour	Work
0:00-0:30	Run `05_STARTER/mock-api` (`node server.ts`). Create Android project with the design system and official logo (Phase 03 steps 2-4 only) and a Vite admin app with the tokens
0:30-1:30	Patient: splash, login (demo account), home with the All-in-One grid (4 tiles)
1:30-2:30	Booking flow: pick service, professional list, confirm, sandbox pay, status screen
2:30-3:30	Health Passport: PIN set/verify (lock after 5 tries), passport view, grant to booking
3:30-4:15	SOS: hold-to-send button, SOS active screen, Admin red alert banner via SSE
4:15-5:00	Polish: states, copy, icons, run DEMO_SCRIPT.md twice, record backup video

Prompt to give the AI

"Run the Hackathon 5H plan. Use the mock API at http://10.0.2.2:4000/v1. Build only showcase screens. Use the design system and official logo from the kit. No emoji. Follow SOS_SPEC.md honesty rules: label anything not real as prototype. Stop after each hour block and report."

Honesty note for judges

State plainly: payments are sandbox, ambulance dispatch is not simulated, and SOS reaches the admin dashboard and emergency contact only.

13. QA checklist and demo script

Functional

- [ ] Signup, OTP, login, logout, refresh rotation
- [ ] Professional: submit documents, see verification status, admin approves, appears in search
- [ ] Booking: all four types (doctor appointment, nurse home visit, medicine order, lab test) → pay (sandbox) → accept → track → complete → rate
- [ ] Duplicate accept returns 409; duplicate create with same Idempotency-Key returns same booking
- [ ] Health Passport: PIN set, lock after 5 wrong tries, grant to booking, professional reads, access log shows read, passport expires
- [ ] SOS: create, location updates, admin sees red alert, acknowledge/resolve, false alarm
- [ ] Notifications arrive; deep links open the right screen

Visual

- [ ] Every screen uses tokens only; same colors on Android, admin, website
- [ ] Official logo appears on splash, login, app bar/sidebar, website header, favicon, launcher icon
- [ ] No emoji anywhere (run: grep for emoji ranges in src and strings)
- [ ] 320/360/411 widths, font scale 1.3, RTL Urdu
- [ ] Loading, empty, error and offline states exist on every list screen

Security

- [ ] Items in SECURITY_PRIVACY.md all pass; IDOR tests for booking, document, passport
- [ ] No secrets in repo; `.env.example` complete
- [ ] Dependency audit has no critical issues

Performance and release

- [ ] Cold start under 2.5 s; no ANR; R8 enabled, signed AAB builds
- [ ] Crash reporting configured; backups configured; staging smoke test passes

Demo script (5 minutes)

1. Splash with official logo. Sign in as Ali (patient).
2. Home: All-in-One Booking grid (doctor, nurse, medicine, lab). Book "Nurse home visit" with Sara Khan; pay with sandbox.
3. Switch to Sara's app: accept the request, mark On the way.
4. Patient opens Health Passport: enter PIN 1234, grant access for this booking. Sara opens the passport; patient shows the access log entry.
5. Patient presses and holds SOS. Admin dashboard (second window) flashes the red alert with name, phone, emergency contact and location. Admin acknowledges, then resolves.
6. Admin: verification queue, approve Hina Tariq.
7. Close with the website waitlist page in English and Urdu.

Talking points: native Android, one design system across three surfaces, audit trail, honest emergency behavior, phased roles.

14. Known limits

- Android and backend source code is not included; the AI builds it from these specs. The schema, OpenAPI file and mock API were tested here.
- Assessment questions are SAMPLES and not clinically validated; a licensed clinician must review them.
- Payments and SMS are sandbox/provider interfaces; real credentials come from the owner.
- The official logo is raster 3D art, so use the supplied density files and do not upscale. A vector master is recommended.
- Urdu text in this PDF uses a system fallback font; bundle Noto Nastaliq Urdu in the apps and website.
- Legal and regulatory points (PMDC, PNC, DRAP, data protection) must be confirmed by counsel.